

Roll No.:

END SEMESTER EXAMINATION JUNE 2025

Name of the Program: B.Tech

Semester: VI

Name of the Course: Advanced Machine Learning

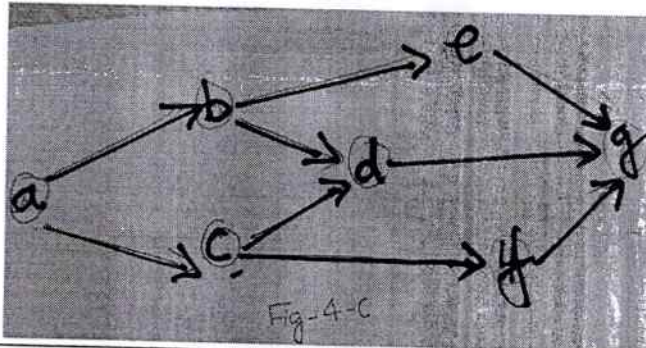
Course Code: TCS 682

Time: 3 Hours

Maximum Marks: 100

Note:

- i. All questions are compulsory.
- ii. Answer any two sub questions among a, b & c in each main questions
- iii. Total marks in each main question are twenty.
- iv. Each questions carries 20 marks

Q1	(20Marks)	
(a)	What are Factor Graphs? How do they relate to graphical models and what are their applications in machine learning?	CO4
(b)	What is k-fold cross-validation? Describe the process with a diagram and explain how it helps in model selection and preventing overfitting.	CO3
(c)	You are given a Medical dataset of heart attack diagnose (Pain, Family History, high cholesterol etc.) and diagnosing heart attack. How would you model this using a Bayesian Network? Explain the steps involved and how you would use it for diagnosing heart attack.	CO3, 4
Q2		
(a)	What are Markov Random Fields? Discuss their factorization properties. Convert Fig- 4-c directed graph into undirected graph. 	CO5
(b)	What is the Bias-Variance tradeoff? Discuss its importance in model evaluation.	CO2
(c)	Given a noisy dataset with missing values, how would you use the Expectation-Maximization (EM) algorithm to estimate the parameters of a Mixture of Gaussians model? Discuss how the EM algorithm can help in clustering the data.	CO3
Q3		
(a)	Explain the concepts of Cross-Validation and its significance in model evaluation.	CO2
(b)	Explain the concepts of Lasso and Maximum Likelihood Estimation (MLE). How do they help in model regularization and estimation?	CO2
(c)	You are given a dataset of customer purchasing behavior, and you want to group customers into similar segments based on their buying patterns. How would you use K-means clustering to achieve this? What challenges might you face in selecting the optimal number of clusters?	CO3, CO6
Q4		
(a)	Explain Classification and Regression with their typical use cases.	CO3

(b)	What are Hidden Markov Models? Explain their applications in machine learning.	CO4
(c)	Consider a simple chain: $A \rightarrow B \rightarrow C$. Are A and C conditionally independent given B? Justify your answer using d-separation.	CO5
Q5		
(a)	Write short Note. 1. Active Learning 2. Online Learning 3. Ensemble Learning	CO5
(b)	Discuss how the concept of KL-Divergence (Kullback-Leibler Divergence) is used to measure the difference between two probability distributions in machine learning.	CO6
(c)	Given a dataset of continuous features, you are asked to estimate the probability density function. How would you implement Kernel Density Estimation (KDE) to estimate the distribution? Discuss the importance of choosing an appropriate bandwidth.	CO6